



Mukund Jangid

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• ABOUT ME

Technical Product & Engineering Leader with 9+ years of experience architecting and delivering distributed, cloud-native platforms across high-compliance domains including FinTech, Life Sciences, and ChemTech. Experienced in taking complex products from zero to enterprise scale, with deep expertise in Agentic AI, machine learning, microservices, Kubernetes, and platform engineering. Inventor of a patent-listed regulatory AI agent that reduced SDS generation time from 2 weeks to 5 minutes. Combines hands-on technical depth with product ownership and an MBA in Analytics to translate complex business and regulatory problems into scalable, resilient technology platforms.

• WORK EXPERIENCE

31/08/2024 - CURRENT - HYDERABAD, INDIA

AI PRODUCT OWNER - AGENTIC APPLICATION DEVELOPMENT VERTEX INC.

- **Technical Product Ownership:** Owned strategy, architecture, and delivery for modernizing a legacy tax platform into a cloud-native, AI-enabled platform.
- **Agentic AI Platform:** Built an API-first agent platform with **LLM agents, guardrails**, and human-in-the-loop controls for enterprise use cases.
- **AI-Driven Modernization:** Developed an LLM-based **Code Analysis Agent** to map legacy Java dependencies, reducing modernization analysis effort by 40%.
- **Domain AI:** Led **Tax Knowledge and Tax Rules Agents** using LLMs, RAG, embeddings, and prompt engineering to analyze tax logic and regulatory requirements.
- **Cloud Architecture:** Architected **Kubernetes-based microservices on Azure**, balancing scalability, compliance, reliability, and cost.
- **Compliance-as-Code:** Productized **OPA-based policy enforcement** for automated security and regulatory compliance.
- **Platform Engineering:** Built Golden Path templates and self-service workflows, reducing service onboarding from days to hours.
- **Data & Reliability:** Led **Flink-based streaming infrastructure** and platform reliability for enterprise-scale workloads.
- **DevOps & Delivery:** Established automated multi-environment deployments using Argo CD and Temporal, with standardized CI/CD and release processes.

Business or sector Information and communication | **Department:** Compliance & Regulations | **Website:** <https://www.vertexinc.com/>

31/07/2022 - 31/08/2024 - BENGALURU, INDIA

TECHNICAL LEAD & FOUNDING ENGINEER ALBERT INVENT

- **Chemical Intelligence Platform:** Owned architecture and delivery of a zero-to-one platform for chemical intelligence and regulatory compliance.
- **Chemist Agent:** Led development of a patented AI system that generates **GHS-compliant SDSs**, reducing generation time from 2 weeks to 15 minutes.
- **Machine Learning:** Built ML-driven workflows using embeddings, retrieval, and chemical/regulatory knowledge bases to analyze properties and jurisdiction-specific rules.
- **Regulatory Intelligence:** Worked with chemists and regulatory experts to convert complex domain knowledge into automated compliance workflows with validation and human review.
- **Platform Architecture:** Designed a **graph-based backend** for chemical and regulatory relationships,

enabling sub-second compliance checks.

- **Kubernetes Platform:** Developed **Go/Python Kubernetes controllers and operators** for environment provisioning, access control, sandboxing, and resets.
- **ML Infrastructure:** Deployed microservices and ML workloads on Kubernetes using **Knative and Istio** for autoscaling and service networking.
- **Enterprise Readiness:** Established **CI/CD, security controls, and SOC 2 readiness**, taking the product from prototype to enterprise deployments across China and Taiwan.

Business or sector Information and communication | **Department:** Regulatory and Compliance Automation | **Website:** <https://www.albertinvent.com/>

31/05/2020 - 30/11/2021 - PUNE, INDIA

R&D ENGINEER - LIFE SCIENCES / DIGITAL TWINDASSAULT SYSTÈMES (MEDIDATA)

- **Digital Human Twin Platform:** Developed a medical sensor data platform integrating enterprise wearables and building digital human twin profiles.
- **IoT & Edge Computing:** Designed high-throughput **edge-to-cloud ingestion** for real-time medical sensor telemetry.
- **Data Engineering:** Built fault-tolerant **Kafka pipelines** for sensitive patient data supporting decentralized clinical trials.
- **CI/CD:** Built and maintained **Jenkins pipelines** for continuous integration and deployment.
- **Automation:** Automated system configuration and environment management using **Puppet and Bash**.
- **Monitoring:** Designed **Grafana dashboards and alerts** for service health, performance, and resource utilization.

Business or sector Information and communication | **Department:** Research and Development - Sensor Data Platform | **Website:** <https://www.medidata.com/>

30/04/2017 - 31/05/2020 - PUNE, INDIA

SENIOR CONSULTANT - FULL STACK ENGINEERCAPGEMINI (MORGAN STANLEY)

- **Enterprise Data Engineering:** Developed data ingestion, migration, and ETL-based event pipelines for enterprise clients.
- **Financial Data Integrity:** Built reusable data synchronization tools for financial data integrity and reconciliation across systems.
- **Low-Latency Systems:** Developed **Java/Scala trading and settlement applications**, optimizing concurrency and garbage collection for microsecond-level latency requirements.
- **FinTech Engineering:** Worked in a high-reliability investment banking environment with rigorous testing, regression automation, and defensive coding practices.
- **Data Governance:** Partnered with governance teams to support **CCPA and data privacy compliance**.

Business or sector Information and communication | **Department:** Audit and Compliance | **Website:** <https://www.morganstanley.com/>

● **EDUCATION & TRAINING**

31/03/2022 - 30/04/2024 - KOZHIKODE, INDIA

EXECUTIVE MBA IN SUPPLY CHAIN & BUSINESS ANALYTICS- INDIAN INSTITUTE OF MANAGEMENT

Final grade: GPA 3.7 / 5.0 | **Level in EQF:** 7

31/05/2013 - 31/07/2017 - JAIPUR, INDIA

BACHELOR OF TECHNOLOGY (B.TECH.) IN COMPUTER SCIENCE- SWAMI KESHVANAND INSTITUTE OF TECHNOLOGY (SKIT)

Final grade: GPA 8.2 / 10 | **Level in EQF:** 6

● **SKILLS**

Microservices | Event-Driven Architecture | SOLID Principles | Domain-Driven Design (DDD) | Strangler Fig

Pattern | Serverless | Ollama | Azure Foundry | LangChain | Multi-Agent Orchestration | LangGraph | LangSmith | Microsoft Agent Framework | Agentic Workflow Design | Model Context Protocol (MCP) | Tool / Function Calling | Prompt Engineering | LLM Evaluation and Observability | Agent Memory and State Management | Human-in-the-Loop Agent Design | RAG (Retrieval-Augmented Generation) | Vector Databases (Pinecone, ChromaDB) | OpenAI API | Hugging Face | .NET Core | C# | Java (17+) | Spring Boot | Python | Golang | React | JavaScript | Node.js | SQL (PostgreSQL, AWS Aurora) | NoSQL (MongoDB, DynamoDB) | AWS | Microsoft Azure | Azure Kubernetes Service (AKS) | GCP (Google Cloud Platform) | Kubernetes | Docker | Terraform | CI/CD (Jenkins, GitHub Actions, Argo CD)

● PROJECTS

Sales and Use Tax Compliance Platform (Vertex Cloud Platform)

Outcome: Enabled faster, safer modernization of a legacy tax platform while reducing compliance risk and developer friction.

- Defined the **product vision and success metrics** for transitioning a monolithic tax system into a compliant, cloud-native platform.
- Identified key customer (internal developer & platform team) pain points around release velocity, regulatory risk, and environment consistency.
- Shaped the platform capability roadmap covering self-service onboarding, compliance automation, and multi-environment parity.
- Drove prioritization decisions balancing modernization speed vs. regulatory assurance in a high-compliance domain.
- Partnered with security, compliance, and engineering stakeholders to validate platform capabilities before broader adoption.

Regulatory & Compliance - Automated SDS

Outcome: Reduced regulatory turnaround time from weeks to minutes for chemical R&D teams.

- Defined the problem statement, success metrics, and MVP scope for automated SDS generation in regulated environments.
- Converted complex regulatory rules into productized workflows consumable by enterprise users.
- Designed multi-tenant sandboxing as a product capability to safely test regulatory scenarios without production risk.
- Partnered with compliance and domain experts to validate outputs and ensure regulatory correctness.

Genomics Research Platform

Outcome: Enabled self-service cohort creation and faster study execution for genomic researchers.

- Translated researcher workflows into user-facing product features for cohort definition, execution, and analysis.
- Defined access control, data isolation, and auditability requirements for regulated research use cases.
- Established product KPIs around study setup time, query performance, and usability.
- Collaborated with engineering teams to sequence delivery across data ingestion, analytics, and UI capabilities.

Sensor Cloud Data Fabric (Digital Human Twin)

Outcome: Supported near real-time monitoring of patient vitals in decentralized clinical trials.

- Contributed to defining the long-term product vision for a unified sensor data and clinical analytics platform.
- Identified core platform capabilities required for real-time ingestion, monitoring, and observability.
- Supported rollout of operational dashboards and data quality metrics for clinical trial teams.
- Ensured alignment with healthcare compliance and data integrity expectations.

CCPA Compliance & Data Modernization

Outcome: Enabled enterprise readiness for evolving data privacy regulations.

- Helped define **compliance-driven requirements** for data migration and audit workflows.
- Identified risk areas related to data access, retention, and traceability.
- Supported delivery of privacy-aligned data handling patterns in regulated enterprise systems.

● **PATENTS AND VENTURES**

USA and Europe Listed Patent - System and Method for Automated Safety Data Sheet Generation

System and Method for Automated Safety Data Sheet Generation (AI-driven method of generating regulatory outputs); WIPO PCT listed, with national-phase publications in Europe, Japan and China.

Links <https://patents.google.com/patent/WO2024233807A1> |

<https://register.epo.org/application?number=EP24804282&tab=main>

Form automation & Management - FormsAI

A registry for every form your business touches. Extract field schemas, map your data records, run submissions at volume, and route exceptions to human operators.

- Automatic field-schema extraction from live portals and forms
- Mapping layer between internal database records and target fields
- Bulk submission pipeline with retries, validation, and idempotency
- Schema-drift detection when target portals change layout
- Shared exception queue with one-click human review

Agent observability & guardrails

Trace every model call, tool execution, and token cost across your agent fleet in real-time. Trip guardrail policies mid-run to block PII leaks, prompt injections, and cost spikes.

- Step-level traces across tool calls, context windows, and model decisions
- Active guardrails that inspect payload inputs & outputs to redact PII and block injections
- Loop detection and hard budget circuit breakers with automatic execution halt
- Real-time token usage, latency, and cost attribution per agent & customer

Page agent — crawl, extract, fill, submit

A Playwright-driven agent that operates real websites like a person: navigates, reads the DOM, crawls for missing information, fills forms, and submits.

- Vision + DOM reasoning that operates without custom APIs
- Crawls sibling sources when required values are missing in payload
- Handles multi-step wizards, file uploads, dropdowns, and CAPTCHAs
- Self-healing selectors that survive portal UI changes
- Every run replays as a video recording and Playwright trace

AI voice agents

Voice agents that hold natural conversations — sub-second response times, live mid-sentence tool lookups, barge-in handling, and clean CRM writebacks.

- ~380ms median response with true natural interruption handling
- Live retrieval and real-time database queries during phone calls
- Warm transfer to human agents with full conversation context
- Automatic call summaries, dispositions, and CRM updates
- Every call transcribed, sentiment scored, and searchable

● **ADDITIONAL INFORMATION**

References

Available upon request.